



Lab Sheet 5

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3 Semester (Session 2026-2025)

Objective:

To understand and practice SQL Joins.

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 5: SQL Joins

Use suitable tables such as Student, Department, Course, and Employee to perform the following:

1. Demonstrate an **INNER JOIN** between Student and Department.

```
156 • SELECT * FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	Kashpur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
106	Maya	Female	20	Rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai

2. Display student names along with their department names using INNER JOIN.

```
157 • SELECT Student.Name, Department.Department_Name FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID;
```

Name	Department_Name
Anshika	IT
Radha	IT
Jayshree	IT
Ruhani	IT
Shreya	IT
Maya	HR
Rahul	HR
Priya	HR
Aman	HR
Neha	HR
Rohit	Finance
Simran	Finance
Aditi	Finance

3. Display employee names along with their department names.

```
158 • SELECT Employee.Name, Department.Department_Name FROM Employee INNER JOIN Department  
159 ON Employee.Department = Department.Department_Name;
```

Name	Department_Name
Priya	IT
Amit	IT
Rohit	HR
Neha	Finance
Rahul	Marketing

4. Display students along with their course details.

```
160 • SELECT Student.Name, Student.Course, Course.Course_ID, Course.Fees, Course.Duration  
161 FROM Student INNER JOIN Course ON Student.Course = Course.Course_Name;
```

Name	Course	Course_ID	Fees	Duration
Anshika	B.Tech CSE	1	90000.00	4 Years
Radha	B.Tech IT	3	80000.00	4 Years
Jayshree	B.Tech IT	3	80000.00	4 Years
Ruhani	B.Tech CSE	1	90000.00	4 Years
Shreya	B.Tech CSE	1	90000.00	4 Years
Maya	B.Tech CSE	1	90000.00	4 Years
Rahul	B.Tech CSE	1	90000.00	4 Years
Priya	BCA	2	60000.00	3 Years
Aman	B.Tech IT	3	80000.00	4 Years
Neha	B.Tech CSE	1	90000.00	4 Years
Rohit	BCA	2	60000.00	3 Years
Simran	B.Tech IT	3	80000.00	4 Years
Aditi	B.Tech CSE	1	90000.00	4 Years
Arjun	BCA	2	60000.00	3 Years

5. Find students belonging to a particular department using INNER JOIN.

```

162 • SELECT Student.Name, Department.Department_Name FROM Student
163 INNER JOIN Department ON Student.Department_ID = Department.Department_ID
164 WHERE Department.Department_Name = 'IT';

```

Name	Department_Name
Anshika	IT
Radha	IT
Jayshree	IT
Ruhani	IT
Shreya	IT

6. Demonstrate a **LEFT OUTER JOIN** between Student and Department

```

165 • SELECT * FROM Student LEFT JOIN Department
166 ON Student.Department_ID = Department.Department_ID;

```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
112	Simran	Female	22	Anritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai
141	Arjun	Male	21	Delhi	B.Tech CSE	60.00	9876543290	103	103	Finance	Mumbai

7. Display all students, including those who do not have a matching department.

```

167 • SELECT Student.Name, Department.Department_Name
168 FROM Student LEFT JOIN Department ON Student.Department_ID = Department.Department_ID;

```

Name	Department_Name
Anshika	IT
Radha	IT
Jayshree	IT
Ruhani	IT
Shreya	IT
Maya	HR
Rahul	HR
Priya	HR
Aman	HR
Neha	HR
Rohit	Finance
Simran	Finance
Aditi	Finance
Arjun	Finance

8. Demonstrate a **RIGHT OUTER JOIN** between Student and Department.

```

169 • SELECT * FROM Student RIGHT JOIN Department ON Student.Department_ID = Department.Department_ID;

```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai
112	Simran	Female	22	Anritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
1003								104	104	Marketing	Pune
1004								105	105	Marketing	Pune

9. Display all departments, including departments having no students.

```

170 • SELECT Department.Department_Name, Student.Name FROM Student
171 RIGHT JOIN Department ON Student.Department_ID = Department.Department_ID;

```

Department_Name	Name
IT	Shreya
IT	Ruhani
IT	Jayshree
IT	Radha
IT	Anshika
HR	Neha
HR	Aman
HR	Priya
HR	Rahul
HR	Maya
Finance	Aditi
Finance	Simran
Finance	Rohit
Marketing	
Marketing	

10. Find students who are not assigned to any department using LEFT JOIN.

```

172 • SELECT Student.Student_ID, Student.Name FROM Student LEFT JOIN Department ON
173 Student.Department_ID = Department.Department_ID
174 WHERE Department.Department_ID IS NULL;

```

Student_ID	Name
141	Arjun

12. Demonstrate a **FULL OUTER JOIN** concept using suitable queries/workaround where supported.

```

175 • SELECT Student.Student_ID, Student.Name, Department.Department_Name
176 FROM Student LEFT JOIN Department ON Student.Department_ID = Department.Department_ID
177 UNION SELECT Student.Student_ID, Student.Name, Department.Department_Name
178 FROM Student RIGHT JOIN Department ON Student.Department_ID = Department.Department_ID;

```

Student_ID	Name	Department_Name
101	Anshika	IT
102	Radha	IT
103	Jayshree	IT
104	Ruhani	IT
105	Shreya	IT
106	Maya	HR
107	Rahul	HR
108	Priya	HR
109	Aman	HR
110	Neha	HR
111	Rohit	Finance
112	Simran	Finance
114	Aditi	Finance
141	Arjun	Marketing

13. Demonstrate a **CROSS JOIN** between Student and Course.

```

179 • SELECT Student.Name, Course.Course_Name FROM Student CROSS JOIN Course

```

Name	Course_Name
Anshika	MBA
Anshika	MCA
Anshika	B.Tech IT
Anshika	BCA
Anshika	B.Tech CSE
Radha	MBA
Radha	MCA
Radha	B.Tech IT
Radha	BCA
Radha	B.Tech CSE
Jayshree	MBA
Jayshree	MCA
Jayshree	B.Tech IT
Jayshree	BCA
Jayshree	B.Tech CSE

14. Calculate the number of possible Student-Course combinations using CROSS JOIN.

```

180 • SELECT COUNT(*) AS Total_Combinations FROM Student CROSS JOIN Course;

```

Total_Combinations
70

15. Demonstrate a **NATURAL JOIN** between two suitable tables.

```

181 • SELECT * FROM Student NATURAL JOIN Department;

```

Department_ID	Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_Name	Location
101	101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	IT	Delhi
101	102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	IT	Delhi
101	103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	IT	Delhi
101	104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	IT	Delhi
101	105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	IT	Delhi
102	106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	HR	Jaipur
102	107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	HR	Jaipur
102	108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	HR	Jaipur
102	109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	HR	Jaipur
102	110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	HR	Jaipur
103	111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	Finance	Mumbai
103	112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	Finance	Mumbai
103	114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	Finance	Mumbai

16. Compare the result of NATURAL JOIN and INNER JOIN.

```

181 • SELECT * FROM Student NATURAL JOIN Department;
182 • SELECT * FROM Student
183 INNER JOIN Department ON Student.Department_ID = Department.Department_ID;

```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Department_ID	Department_ID	Department_Name	Location
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210	101	101	IT	Delhi
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276	101	101	IT	Delhi
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	101	101	IT	Delhi
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	102	102	HR	Jaipur
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	102	102	HR	Jaipur
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	102	102	HR	Jaipur
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	102	102	HR	Jaipur
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	102	102	HR	Jaipur
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	103	103	Finance	Mumbai
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	103	103	Finance	Mumbai
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250	103	103	Finance	Mumbai

17. Perform a **SELF JOIN** on an Employee table to show employee-manager relationships.

```

191 • SELECT E.Name AS Employee, M.Name AS Manager
192 FROM Employee E LEFT JOIN Employee M ON E.Manager_ID = M.Employee_ID;

```

Employee	Manager
Amit	NULL
Rahul	Amit
Neha	Amit
Priya	Rahul
Rohit	Rahul

19. Join three tables—Student, Department and Course.

```

194 FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID
195 INNER JOIN Course ON Student.Course = Course.Course_Name;

```

Student_Name	Course_Name	Department_Name
Anshika	B.Tech CSE	IT
Radha	B.Tech IT	IT
Jayshree	B.Tech IT	IT
Ruhani	B.Tech CSE	IT
Shreya	B.Tech CSE	IT
Maya	B.Tech CSE	HR
Rahul	B.Tech CSE	HR
Priya	BCA	HR
Aman	B.Tech IT	HR
Neha	B.Tech CSE	HR
Rohit	BCA	Finance
Simran	B.Tech IT	Finance

20. Display student name, course name and department name using multiple joins.

```

196 SELECT Student.Name AS Student_Name, Course.Course_Name, Department.Department_Name
197 FROM Student INNER JOIN Department ON Student.Department_ID = Department.Department_ID
198 INNER JOIN Course ON Student.Course = Course.Course_Name;

```

Student_Name	Course_Name	Department_Name
Anshika	B.Tech CSE	IT
Radha	B.Tech IT	IT
Jayshree	B.Tech IT	IT
Ruhani	B.Tech CSE	IT
Shreya	B.Tech CSE	IT
Maya	B.Tech CSE	HR
Rahul	B.Tech CSE	HR
Priya	BCA	HR
Aman	B.Tech IT	HR
Neha	B.Tech CSE	HR
Rohit	BCA	Finance
Simran	B.Tech IT	Finance
Aditi	B.Tech CSE	Finance

21. Find employees whose salary is greater than the average salary of their department using joins/subqueries.

```

199 SELECT E.Name, E.Department, E.Salary FROM Employee E
200 WHERE E.Salary > (SELECT AVG(E2.Salary) FROM Employee E2 WHERE E2.Department = E.Department);

```

Name	Department	Salary
Priva	IT	45980.00

22. Find the department having the maximum number of students using joins and aggregation.

```

201 SELECT Department.Department_Name, COUNT(Student.Student_ID) AS Total_Students
202 FROM Department LEFT JOIN Student ON Department.Department_ID = Student.Department_ID
203 GROUP BY Department.Department_ID, Department.Department_Name ORDER BY Total_Students DESC LIMIT 1;

```

Department_Name	Total_Students
IT	5

23. Display departments and the total number of students in each department.

```

204 SELECT Department.Department_Name, COUNT(Student.Student_ID) AS Total_Students
205 FROM Department LEFT JOIN Student ON Department.Department_ID = Student.Department_ID
206 GROUP BY Department.Department_ID, Department.Department_Name;

```

Department_Name	Total_Students
IT	5
HR	5
Finance	3
Marketing	0
Sales	0

24. Display courses along with the number of students enrolled in each course.

```

207 SELECT Course.Course_Name, COUNT(Student.Student_ID) AS Total_Students
208 FROM Course LEFT JOIN Student ON Student.Course = Course.Course_Name
209 GROUP BY Course.Course_ID, Course.Course_Name;

```

Course_Name	Total_Students
-------------	----------------

25. Create a complete query using **INNER JOIN**, **LEFT JOIN**, **GROUP BY** and **aggregate functions** to generate a student-department-course report.

```
210 • SELECT Department.Department_Name, Course.Course_Name,  
211 COUNT(Student.Student_ID) AS Total_Students,  
212 AVG(Student.Percentage) AS Average_Percentage  
213 FROM Department LEFT JOIN Student ON Department.Department_ID = Student.Department_ID LEFT JOIN Course  
214 ON Student.Course = Course.Course_Name GROUP BY  
215 Department.Department_ID,Department.Department_Name, Course.Course_ID,Course.Course_Name;
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Contents: I5

Department_Name	Course_Name	Total_Students	Average_Percentage
IT	B.Tech CSE	3	85.500000
IT	B.Tech IT	2	85.500000
HR	B.Tech CSE	3	84.083333
HR	B.Tech IT	1	74.750000
HR	BCA	1	91.000000
Finance	B.Tech CSE	1	90.500000
Finance	B.Tech IT	1	82.000000
Finance	BCA	1	69.250000
Marketing	None	0	None
Other	None	0	None

Submission:

Take screenshots of each step's output.

Save the SQL scripts in a text file.